Unlocking households' flexibility potential - on the barriers to real-time pricing*

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Abstract

Many countries intend to electrify most of their energy consumption in the near future, which can raise the flexibility of electricity consumption significantly. To unlock this flexibility potential, consumers require appropriate price signals, achieved through real-time pricing (RTP) contracts. Building on the stylized market model of Borenstein and Holland (2005), this paper identifies potential barriers to the adoption of RTP. We find substantial synergies between RTP and flexibility of consumption. As flexibility increases, so does the adoption of RTP. However, this relationship weakens if consumers' electricity bills are dominated by non-energy components such as grid fees, taxes, and retail markups. When these costs are substantial, RTP becomes less attractive and adoption remains limited.

JEL-Classification: D10, D40, L10, L94, Q41

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1 Introduction

The lack of flexibility in electricity markets can cause frequent price spikes, elevated grid management costs, and in severe situations, could lead to rationing and power outages. The likelihood of these risks increases in the share of intermittent renewables, as their generation cannot be dispatched on demand like conventional power plants. Consequently, a significant boost in flexible capacity will be necessary to handle the energy transition (e.g. Joskow, 2012; IEA, 2016). Countries' ambitions to electrify the majority of their energy consumption may unlock substantial flexibility potential in the near future. For instance, Germany's policy objectives suggest that the capacity for flexible residential consumption will rise to 200 GW by 2030, which is approximately 280% of the current capacity of flexible power plants in Germany (Hirth et al., 2023).[1]

However, residential consumers are only incentivized to utilize this flexibility potential when appropriate price signals exist. At present, most consumers have fixed retail prices and lack incentives to adjust their consumption according to existing supply conditions. Passing on wholesale prices, which mirror contemporaneous supply and demand conditions, could provide such signals and is commonly known as real-time pricing (RTP). It promotes consumption when electricity is produced by low-marginal cost renewables and discourages consumption when it is produced by costly and polluting power plants (Holland and Mansur, 2008; Ambec and Crampes, 2021). Hence, many policymakers and economists consider the widespread adoption of RTP to be a key element in the energy transition.[2]

Despite efforts by policymakers and technological requirements being met, the adoption rate of RTP contracts remains minimal in most European countries (ACER-CEER, 2024). This paper examines the barriers to RTP adoption. To do so, we extend the stylized market model of Borenstein and Holland (2005) along three dimensions: we (i) introduce heterogenous utility-maximizing consumers with intertemporal flexibility in their electricity demand, (ii) endogenize consumers' retail contract choice, and (iii) account for frictions in the retail market.

Put together, our model reveals nontrivial mechanism that can limit the adoption of RTP contracts by consumers. First, our results indicate notable synergies between the flexibility of consumption and RTP. If flexibility is low, consumers cannot easily adjust their consumption according to prices, rendering RTP unattractive. However, as flexibility rises, consumers can more effectively shift consumption across time, increasing consumers' benefits of RTP contracts and thus adoption.

[1]To put the number in perspective: as of 2024, the government of Germany aims to build 10 GW additional gas power plant capacity by 2030 to improve the flexibility of the electricity market. See website of German Parliament (last accessed: January 2026).

[2]As such, a number of European countries mandated retailers to offer a RTP contract option to consumers as required by EU legislation (European Union, 2024).

Such synergies may however disappear if frictions in the retail market are significant, i.e., if wholesale and retail prices are highly dispersed. In this paper, we consider two distortions commonly found in practice: (i) lump-sum fees/taxes and (ii) market power. If these distortions are significant, which empirical evidence suggests (ACER-CEER, 2024), consumer savings from RTP remain negligible and adoption limited. This result is driven by equilibrium effects induced by the higher elasticity of demand during peak periods. As markups raise the average retail prices for consumers, demand in peak periods decreases by relatively more than in off-peak periods, which leads to a decrease in the real-time price volatility as off-peak prices increase by relatively more than peak prices. Since RTP functions as an 'arbitrage opportunity' to exploit existing price discrepancies, a reduction in these discrepancies lowers the attractiveness of RTP to consumers.

Our results are computed for a specific parametric framework but remain robust as long as (i) the industry marginal cost of generating electricity is non-concave and (ii) demand in peak periods is relatively more elastic than in off-peak periods; both common assumptions in the peak-load pricing literature (Crew et al., 1995).

Our study builds on the theoretical literature on real-time pricing, going back to Borenstein and Holland (2005) and Joskow and Tirole (2006); the frameworks of which have been extended and applied to different contexts (e.g. Borenstein, 2005; Holland and Mansur, 2006, 2008; Savolainen and Svento, 2012; Gambardella et al., 2020; Poletti and Wright, 2020). A common feature of this literature is the exogenous treatment of RTP adoption, in which the first-best outcome is achieved when all consumers adopt RTP contracts. Given the low adoption rate of these contracts in practice (e.g. Pébereau and Remmy, 2023; ACER-CEER, 2024), we contribute to the literature by considering heterogeneous consumers, endogenizing consumers' contract choices, and identifying barriers that prevent the socially optimal outcome of full RTP adoption.

This paper also relates to the empirical literature investigating the flexibility of electricity consumption across time. Utilizing natural experiment settings, a number of studies find that households' responses to time-varying electricity prices remain limited, resulting in minimal benefits for consumers (e.g. Wolak, 2011; Allcott, 2011; Ito, 2014; Harding and Sexton, 2017; Fabra et al., 2021). On the other hand, empirical evidence also suggests substantial savings potential among more responsive consumers, such as commercial consumers (Patrick and Wolak, 2001; Borenstein, 2007) or households adopting automation technologies (Bollinger and Hartmann, 2020; Bailey et al., 2025). We contribute to the empirical literature by proposing a parsimonious demand framework, that models electricity consumption at different times as substitutable. Our model setup allows for different degrees of intertemporal flexibility of demand and confirms empirical intuition.

Empirical evidence also suggests that inflexibility in electricity consumption stems from final consumers potentially not behaving rational, which in turn influences their retailer contract choice as well, e.g., through risk aversion (Boom and Schwenen, 2021) or inertia and limited awareness (e.g. Hortaçsu et al., 2017; Fowlie et al., 2021; Dressler and Weiergraeber, 2023; Pébereau and Remmy, 2023). Our model also investigates contract choice and suggests that even rational and flexible consumers—absent of any behavioral biases— may have no incentive to adopt RTP if certain retail market conditions prevail. For example, if retailers have market power, markups charged to consumers mute the relative volatility of retail prices across time-periods, limiting the attractiveness of RTP.

Lastly, our paper contributes to the debate on grid-related fees and taxes in the final electricity bill of consumers. In most markets, a significant share of the final consumer bill is spent on grid fees and taxes, while the procurement of electricity accounts only for roughly half of the bill (ACER-CEER, 2024). As such, there is an ongoing debate discussing the introduction of time-varying grid fees to effectively account for grid conditions (Stute and Klobasa, 2024). According to our model, such a policy change would benefit the adoption of RTP. Similar to markups from market power, lump-sum fees and taxes mute differences in real-time retail prices, lowering the potential gains of RTP for consumers. However, if fees are charged time-varying or proportionally to the wholesale price, the additional volatility in retail prices may induce the adoption of RTP.

2 Model Setup

This section introduces the general model setup and the utility measure of consumers that generates intertemporal demand flexibility. We follow the setup of Borenstein and Holland (2005). Each day consists of $t \in \mathcal{T}$ time periods, in which consumers demand electricity and producers generate it. We index time periods as $t \in (1, \dots, T)$, which we simplify to a stylized two-period model later on. In that sense, time periods can be hours or larger time blocks, such as daytime and nighttime.

Our electricity market model consists of a wholesale and retail market with producers, retailers, and consumers. On the wholesale market, producers sell electricity to retailers. Simultaneously, retailers distribute the purchased electricity to consumers in the retail market.

Producers sell electricity to retailers for wholesale price w_t in every period t. Each producer is relatively small compared to the market and has an identical production technology with increasing marginal cost. The industry variable cost of production is given by $C(q)$, where q is the quantity generated. The industry cost function is characterized by increasing marginal cost, i.e., $C_q > 0$ and $C_{qq} > 0$.[3] The objective function of producers

[3]We omit capacity investment for producers. Similar to Boom and Schwenen (2021), we thus focus our analysis on short-run dynamics.

is given by

$$\pi^w = \sum_{t=0}^{T} w_t q_t - C(q_t). \tag{1}$$

Since each producer is small and has increasing marginal cost, each producer generates the same quantity (Borenstein and Holland, 2005). As such, firm-specific profits will be a constant fraction of sector profits, and the solution that maximizes individual producer profits is identical to the solution that maximizes sector profits.

Retailers buy electricity from producers for price w_t and sell it to consumers under different contracts. Retailers offer two types of contracts. For the real-time pricing (RTP) contract, retailers sell electricity for time-varying prices p_t to consumers. Alternatively, they can offer a fixed contract charging the same price $\bar{p}$ in all periods t. The share of consumers on real-time pricing contracts is denoted by α. For now, we disregard any frictions on the retail market and assume there are many identical retailers which compete à la Bertrand. The objective function of the retail sector is given by

$$\pi^r = \alpha \sum_{t=0}^{T} (p_t - w_t) q_t(p_1, \dots, p_T) + (1-\alpha) \sum_{t=0}^{T} [(\bar{p} - w_t) q_t(\bar{p}), \tag{2}$$

where $q_t(p_1, \dots, p_T)$ is the quantity demanded by real-time pricing consumers in t. $q_t(\bar{p})$ is short-hand notation for $q_t(\bar{p}, \dots, \bar{p})$, the quantity demanded by consumers on a fixed pricing contract in period t, for which the price in every period is equal to $\bar{p}$. Since retailers are identical, the solution that maximizes sector profits is identical to the solution that maximizes individual retailer profits.

Consumers demand electricity in every period t. There is a unit mass of consumers. We assume the utility measure to be of separable form

$$\tilde{U}(q_1, \dots, q_T, M) = U(q_1, \dots, q_T) + M, \tag{3}$$

where $U(q_1, \dots, q_T)$ is utility derived from electricity consumption and M is the consumption of the outside good. Assume U to be twice-differentiable and strictly quasi-concave, which implies that electricity at different times are substitutes (Bergstrom and MacKie-Mason, 1991). Assume M to be the numeraire and that the budget constraint can be written as $M = Y - \sum_t p_t q_t$ with Y being income. Linear separability is a common assumption for preferences in the peak-load pricing literature including utility-maximizing consumers (see Crew et al., 1995). It is particularly useful because it disregards any income effects and allows for a partial equilibrium analysis (Symeonidis, 2003). For this to be true, consumers should only spend a small share of their income on electricity expenditure, which is largely in line with empirical observations.[4]

By assuming electricity at different times to be substitutes, we explicitly model electricity consumption as flexible across time. To do so analytically, we specify the general utility function (3) as the widely-used quality-augmented quadratic utility function, first introduced by Sutton (1997) and Sutton (2001). Specifically,

$$\tilde{U}(q_1,\ldots,q_T,M) = U(q_1,\ldots,q_T) + M = \sum_{t=1}^{T}\left(aq_t - \frac{q_t^2}{d_t^2}\right) - \sigma\sum_{t=1}^{T}\sum_{s<t}^{T}\frac{q_tq_s}{d_td_s} + M, \quad (4)$$

where a is a utility constant, σ captures the degree of substitutability (horizontal differentiation) and d_t the quality of electricity in period t (vertical differentiation). The higher d_t, the larger the utility received by the same amount of consumption in period t. Strictly speaking, electricity is a homogeneous good without quality differences at different times. Instead, one can think of quality as the value consumers attach to the consumption of electricity at different times. For example, consumers value electricity more when they are home than when they are away or when it is hot rather than cold.

Moreover, the main parameter of interest, σ, captures the degree of substitutability between electricity consumption at different times and can range between 0 and 2. For $\sigma = 2$, electricity consumption throughout the day would be perfect substitutes, while for $\sigma = 0$, electricity consumption at different times is independent of each other. The higher σ, the more substitutable electricity becomes across time. This implies that consumers become more responsive to prices in other time periods and hence the more flexible in shifting electricity consumption between time periods. For that reason, we use the terms flexibility and substitutability interchangeably throughout the paper.

This parametric form has multiple advantages. First, it yields a linear demand function, which allows us to solve for the equilibrium as a system of linear equations. Secondly, it is relatively parsimonious with σ capturing the extent of flexibility across time periods and the spread between $(d_1,\ldots,d_T)$ capturing consumers' preferences between time periods.

2.1 Two-period equilibrium

We derive the equilibrium of the model for a stylized two-period setting. This simplification considerably eases underlying derivations and is sufficient to illustrate the general dynamics of our model. For the remainder of the paper, we assume a two-period model with an *off-peak* period (low demand, l) and a *peak* period (high demand, h), $\mathcal{T} = \{l, h\}$. We still consider a daily setting, so l and h could be two time blocks that split one day

[4] The IEA (2023) estimates that home energy expenditure ranges between 3% to 8% of household income in most developed countries in 2021-22. This also includes gas and thus serves as an upper bound for electricity expenditure. The actual electricity bill is likely to be lower.

into peak and off-peak hours. As a notational remark: if we refer to prices and quantities with subscript l or h (e.g., p_l, q_h), we aim to emphasize properties of the specific time period. If we use subscript t (e.g., p_t, q_t), we make statements that hold for prices and quantities in all time periods $t \in \{l, h\}$.

The equilibrium is identified by producers, retailers, and consumers maximizing their respective objective functions and market clearing. We first derive the demand of consumers given utility function (4). In a two-period model, maximizing utility measure (4) given the budget constraint yields the following demand function for each period t

$$q_t(p_t, p_s) = \frac{2d_t^2(a - p_t) - \sigma d_t d_s(a - p_s)}{4 - \sigma^2}, \quad t, s = l, h, \ s \neq t, \tag{5}$$

where s denotes the time period that is not t. We make the following assumption on the utility parameters. First, we define peak period h as the time period, in which consumers have an inherent preference to consume. In other words, the 'quality' parameters are set such that $d_h > d_l$. Consumers therefore value consumption during the peak period more than during the off-peak period. Second, we assume that consumers cannot consume negative amounts of electricity. To ensure this, we define the upper limit for σ as $\bar{\sigma} = 2\frac{d_l}{d_h}$. Note that if $\sigma > \bar{\sigma}$, consumers paying a fixed price $\bar{p}$ would consume a negative amount of electricity in off-peak period l to afford more electricity in peak period h. We refer to the two assumptions as follows:

Assumption U1 (Preference for peak). *For quality parameters $\{d_h, d_l\}$, it holds that $d_h > d_l$.*

Assumption U2 (Nonnegativity of consumption). *Substitutability parameter σ is bounded by $\sigma \in [0, \bar{\sigma}]$, where $\bar{\sigma} = 2d_l/d_h$.*

Consequently, the total demand of consumers, i.e., both RTP and fixed price consumers, in period t is

$$Q_t(p_t, p_s, \bar{p}) = \alpha q_t(p_t, p_s) + (1 - \alpha) q_t(\bar{p}), \quad t, s = l, h, \ s \neq t, \tag{6}$$

where q_t is the quantity demanded given by equation (5).

For every period t, producers maximize profits (1) when wholesale prices equal marginal costs. Given market clearing, wholesale prices are given by

$$w_t = C_q\big(Q_t(p_t, p_s, \bar{p})\big), \quad t, s = l, h, \ s \neq t, \tag{7}$$

and thus equal to the marginal cost of supplying total quantity demanded.

Lastly, equilibrium retail prices are determined on the retail market, where retailers compete in prices. For RTP contracts, equilibrium retail prices are equal to wholesale prices, which are the marginal cost of retailers. For any other price, retailers could deviate and marginally undercut the price to obtain positive profits. Such deviations are possible until prices reach marginal costs (wholesale prices). A similar logic applies to fixed pricing contracts. Assume all retailers charge $\bar{p} = w_h$. Then, retailers break even in the peak period but gain profits in the off-peak period. One retailer could then marginally undercut this price, gaining more consumers. For each consumer, this retailer still gains profits in l but incurs losses in h. Such deviations are profitable until the per-consumer profits and losses offset each other. Hence, prices are set in a fashion where revenues perfectly cover input costs, i.e., zero profits. The retail market equilibrium conditions are thus given by

$$\begin{aligned} p_t &= w_t, \quad \forall t, \\ \bar{p} &= \frac{w_l q_l(\bar{p}) + w_h q_h(\bar{p})}{q_l(\bar{p}) + q_h(\bar{p})}, \end{aligned} \tag{8}$$

where the second equation follows from rearranging the second term of retail profits (2) and setting it equal to zero. As a result of the retail market equilibrium, wholesale prices are passed through to consumers on RTP contracts. Consumers on a fixed pricing contract pay a demand-weighted average of the wholesale prices in both periods. For now, assume that the share of consumers on real-time pricing contracts α is exogenous and that there are essentially two types of consumers. Later on, we relax this assumption and endogenize the contract choice.

This allows us to identify the equilibrium: For a given share of consumers on RTP contracts $\alpha \in (0, 1]$, and substitutability parameter $\sigma \in [0, \bar{\sigma}]$, the set of equilibrium wholesale prices $\{w_l, w_h\}$ and equilibrium retail prices $\{p_l, p_h, \bar{p}\}$ is determined by the individual and total demand functions (5)-(6), wholesale profit-maximizing condition (7), and the retail zero-profit conditions (8). As equilibrium retail prices p_t and wholesale prices w_t are identical, we summarize either as 'real-time' prices and denote them with p_t.[5] To fix ideas, we describe the characteristics of equilibrium in three Lemmas. Lemma 1 describes the relationship of real-time prices.

Lemma 1. *The equilibrium peak price is greater than the equilibrium off-peak price, $p_h > p_l$, for all $\alpha \in (0, 1]$ and $\sigma \in [0, \bar{\sigma}]$.*

Proof. Suppose $d_h > d_l$ and $p_h \leq p_l$. From increasing marginal cost and short-run wholesale profit-maximizing condition (7) follows $Q_h(p_h, p_l, \bar{p}) \leq Q_l(p_l, p_h, \bar{p})$. However, for $d_h > d_l$ and $p_h \leq p_l$, then $q_h(p_h, p_l) > q_l(p_l, p_h)$ and $q_h(\bar{p}) > q_l(\bar{p})$. Thus, $Q_h(p_h, p_l, \bar{p}) > Q_l(p_l, p_h, \bar{p})$, which is a contradiction. □

Lemma 1 states that peak real-time price p_h is strictly larger than off-peak real-time price p_l. This holds independently of the share of RTP consumers and substitutability

[5] In Section 3.3, we return to the original notation due to frictions in the retail market.

of electricity. This follows from Assumption U1. By setting $d_h > d_l$, demand in the peak period is higher and consequently prices due to increasing marginal cost. Lemma 2 describes the properties of the fixed price.

Lemma 2. *Equilibrium fixed price $\bar{p}$ is bounded by p_l and p_h such that $p_l < \bar{p} \leq p_h$ for all $\alpha \in (0, 1]$ and $\sigma \in [0, \bar{\sigma}]$.*

Proof. Following equation (8), the equilibrium fixed price is the demand-weighted average of real-time prices. Given Lemma 1 and nonnegativity of consumption (Assumption U2), $\bar{p}$ must be bounded between p_l and p_h. For all t, $q_t(\bar{p})$ is strictly positive unless $\sigma = \bar{\sigma}$, then $q_l(\bar{p}) = 0$ and $\bar{p} = p_h$; hence the weak inequality for the upper bound. □

Regardless of the share of RTP consumers and subsititutability of electricity, Lemma 2 states that the fixed retail price lies between the two real-time prices. Since the fixed price is a demand-weighted average and demand is higher in h than in l, $q_h(\bar{p}) > q_l(\bar{p})$, the fixed price is always relatively closer to the peak price than the off-peak price. As substitutability increases, the fixed price moves closer to the peak price until eventually converging to it in the corner case of $\sigma = \bar{\sigma}$.

Given the relationship of retail prices, we can infer the relationship of quantity demanded. Since individual demand at time t decreases in its own price and increases in the other period's price, we obtain Lemma 3.

Lemma 3. *Fixed pricing demand is larger than real-time pricing demand in peak period h and smaller in off-peak period l. Hence, $q_h(\bar{p}) \geq q_h(p_h, p_l)$ and $q_l(\bar{p}) < q_l(p_l, p_h)$ for all $\alpha \in (0, 1]$ and $\sigma \in [0, \bar{\sigma}]$.*

Lemma 3 indicates that fixed pricing consumers consume more during the peak period h and less during the off-peak period l than consumers on RTP. This occurs because these consumers are not exposed to any price variation and are thus not incentivized to adjust their consumption according to market conditions. Since RTP consumers face the wholesale price that equals marginal cost, fixed pricing consumers 'overconsume' in the peak period and 'underconsume' in the off-peak period.

Lemma 3 summarizes the main issue related to fixed retail prices. The fixed price induces distortions in the quantity consumed, which deviates from the first-best outcome. Hence, the first-best outcome is only achieved if all consumers are exposed to real-time prices, i.e., if $\alpha = 1$, which is in line with Borenstein and Holland (2005) and subsequent literature. Introducing flexible consumption therefore does not alter the main properties of the model. Nevertheless, our paper sets a different focus. In Borenstein and Holland (2005), the share of RTP consumers α is treated as exogenous with the central result being the non-attainability of the second-best outcome given the existence of fixed retail prices. In contrast, this paper studies the incentives to adopt RTP and identifies underlying mechanisms that hinder or enable the first-best outcome of full adoption.

3 Model Results

This section presents the main results. In a first step, we investigate how market prices and consumer welfare vary as the share of RTP consumers α and substitutability of electricity σ changes. In a second step, we endogenize RTP adoption and analyze how the adoption rate of RTP is affected by the substitutability of electricity and newly introduced retail market frictions.

The subsequent analysis is based on explicit solutions for equilibrium real-time prices in terms of model parameters, which are determined by solving the equilibrium as a system of linear equations. This requires two additional steps. First, the functional form for the marginal cost curve C_q has not been specified. Therefore, assume C_q to be a linear marginal cost function given by $C_q(q) = \lambda q$, where λ is a positive slope parameter. Secondly, it can be shown that the fixed price $\bar{p}$ is a linear function of real-time prices. Plugging individual demand (5) into the equilibrium fixed price expression (8) yields

$$\bar{p}(p_l, p_h) = \frac{2d_l^2 p_l + 2d_h^2 p_h - \sigma d_l d_h (p_l + p_h)}{2d_l^2 + 2d_h^2 - 2\sigma d_l d_h}, \tag{9}$$

which is linear function of real-time prices and shares the properties of Lemma 2.

Consequently, the equilibrium conditions (5)-(8) form a system of linear equations that can be solved for the price pair $\{p_l, p_h\}$. The price functions, that solve the system of equilibrium equations, are provided in Appendix A (see equations (A1)-(A3)). To simplify expressions and subsequent derivations, we normalize $d_l = 1$ and assume $d_h > 1$. Since we use explicit solutions solely to prove inequalities, the results remain valid even without the normalization as long as Assumption U1 ($d_h > d_l$) holds.

3.1 Exogenous RTP adoption

In this section, we treat the share of RTP consumers α as exogenous and analyze how prices and consumer welfare relate to model parameters. Proposition 1 summarizes our results for prices.

Proposition 1. *In equilibrium, (i) the real-time price difference decreases in the share of RTP consumers α and (ii) stronger for higher substitutability of electricity σ. (iii) The equilibrium fixed price decreases in α and (iv) stronger for higher σ.*

Proof. See proof in appendix. □

Part (i) of Proposition 1 describes the effect of RTP adoption on equilibrium real-time prices. As more consumers adopt RTP contracts, consumers reduce consumption in the peak period and increase it in the off-peak period leading to a partial convergence of real-time prices.

Proposition 1 states that these findings also hold in environments with flexible electricity consumption. This is intuitive as both own and cross-price derivatives pull in the same direction. As consumers switch to RTP contracts, they not only increase their consumption in period l due to lower prices in that period, they also increase it due to higher prices in period h. Hence, substitutability amplifies the extent of consumption shifts between periods leading to a quicker convergence of real-time prices in α as stated in part (ii) of Proposition 1.

Part (iii) states that these consumption adjustments lead to a decrease in the fixed price when α increases, which is stronger when substitutability is high (part (iv)). As RTP consumers shift consumption to the off-peak period, peak prices fall. Since the peak price carries a higher weight in determining the fixed prices, the increase in the off-peak price is insufficient to offset the effect of the decreasing peak price, and in aggregate the fixed price decreases as well. This effect is amplified through higher substitutability in two ways. As RTP consumers become more flexible, they shift more consumption away from the peak and into the off-peak period. Additionally, as the number of RTP consumers grows, the number of fixed-pricing consumers decreases. Because these consumers pay the same price in both periods and have an intrinsic preference to consume in h, an increase in substitutability induces relative consumption shifts in the opposite direction, namely into the peak period.[6] Reducing the number of fixed-pricing consumers hence further alleviates peak prices, especially so if substitutability of electricity is high.

In total, these consumption adjustments result in a lower fixed price as the share of RTP consumers increases. That increased adoption of RTP lowers fixed retail prices is a common result in the RTP literature and is sometimes referred to as a 'free riding' phenomenon.[7] In other words, adopting RTP exerts a positive externality on fixed-pricing consumers, making such contracts relatively more attractive. Our model indicates that the size of this externality increases as consumption of electricity becomes more flexible.

To measure the attractiveness of the respective contracts, we define consumer welfare as the respective utility given the underlying contracts the consumer faces. We consider three different types of consumers: (i) fixed pricing consumers, (ii) RTP consumers and (iii) consumers that switch from fixed to RTP contracts. Plugging the individual demand function (5) into the utility measure (4) yields the consumer welfare in terms of prices. Welfare of RTP consumers is given by

$$\tilde{U}^{RTP}(p_l, p_h) = \frac{d_l^2(a-p_l)^2 + d_h^2(a-p_h)^2 - d_l d_h \sigma (a-p_l)(a-p_h)}{4-\sigma^2} + M, \qquad (10)$$

[6]The demand ratio of fixed-pricing consumers, $q_h(\bar{p})/q_l(\bar{p})$, is given by $(2d_h^2 - \sigma d_l d_h)/(2d_l^2 - \sigma d_l d_h)$ As σ increases, consumers shift their relative consumption into the peak period.

[7]This finding goes back to Theorem 3 of Borenstein and Holland (2005).

similarly the welfare of fixed pricing consumers is equal to

$$\tilde{U}^F(\bar{p},\bar{p}) = \frac{(a-\bar{p})^2(d_l^2 + d_h^2 - d_l d_h \sigma)}{4-\sigma^2} + M. \tag{11}$$

Intuitively, the welfare of both consumer types decreases in their respective prices. However, note that the right-most term in the numerator of equation (10) is maximized when both prices are equal. Hence, given some mean-preserving distribution of real-time prices, the welfare of RTP consumers rises with the volatility of real-time prices. While this also holds when electricity consumption is independent, the magnitude of this effect is particularly large when consumers are flexible across time periods. The higher σ, the easier it becomes for RTP consumers to take advantage of lower prices during off-peak period l and shift their consumption accordingly.

Since we are interested in the adoption rate of RTP, our main variable of interest is consumers' incentive to switch contracts, i.e., the welfare gain of switching from a fixed pricing to a RTP contract. We denote the gain of switching as the simple difference between welfare measures, $\tilde{U}^{RTP}(p_l,p_h) - \tilde{U}^F(\bar{p},\bar{p})$. Deriving the switching gain as a simple difference implicitly assumes that each consumer is very small, so consumers do not consider any changes in prices due to consumption adjustments.[8] In terms of prices, we obtain

$$\begin{aligned}\Delta\tilde{U}^S(p_l,p_h,\bar{p}) = &\frac{d_l^2(\bar{p}^2 - p_l^2) + d_h^2(\bar{p}^2 - p_h^2) - d_l d_h \sigma(\bar{p}^2 - p_l p_h)}{4-\sigma^2} \\ &+ a\frac{2d_l^2(p_l - \bar{p}) + 2d_h^2(p_h - \bar{p}) - d_l d_h \sigma(p_l + p_h - 2\bar{p})}{4-\sigma^2}.\end{aligned} \tag{12}$$

By plugging in the equilibrium fixed price (9), the welfare gain of switchers simplifies to

$$\Delta\tilde{U}_p^S(p_l,p_h) = \frac{d_l^2 d_h^2 (p_h - p_l)^2}{4(d_l^2 + d_h^2 - d_l d_h \sigma)}, \tag{13}$$

which is independent of the fixed price and increases with the real-time price difference. The subscript p follows from the equilibrium fixed price condition (9), which follows from *perfect* competition in the retail market. Later on, we derive the switching gain for alternative retail market frictions, where the subscript denotes the respective friction.

The gain of switching signifies the 'arbitrage opportunity' of RTP contracts by highlighting the welfare benefit from leveraging price variations. When real-time prices have fully converged, fixed and real-time pricing contracts become equivalent, and the gain of switching is nil. Conversely, if real-time prices are highly dispersed, switching to RTP constitutes a substantial welfare gain by adjusting consumption to new prices. This leads us to Proposition 2.

[8] A similar assumption has been made in Borenstein and Holland (2005) to derive Theorem 6.

Proposition 2. *Given perfectly competitive retailers, (i) welfare of fixed pricing consumers increases in the share of RTP consumers, (ii) welfare of RTP consumers may decrease, increase or first decrease and then increase in the share of RTP consumers. (iii) The welfare gain of switchers is always positive, decreases in the share of RTP consumers and increases in the substitutability of electricity.*

Proof. See proof in appendix. □

Part (i) of Proposition 2 simply repeats the positive externality of RTP adoption on fixed pricing consumers discussed above. An increase in RTP consumers reduces the fixed retail price and consumers on fixed pricing contracts are better off.

Part (ii) states that the increase the share of RTP consumers has an ambiguous effect on the welfare of RTP consumers. The direction of the effect is determined by the slope of the marginal cost curve in our model framework. If the marginal cost curve is steep, welfare of RTP consumers decreases in α. The reasoning is as follows: for a steep marginal cost curve, real-time price differences are more significant, resulting in RTP consumers to consume more during period l than h. If $q_l > q_h$, raising α increases the overall demand-weighted price for RTP consumers, reducing their welfare. If real-time prices are close to each other, i.e., the marginal cost curve is flat, RTP consumers consume more during the peak period h. If $q_h > q_l$, an increase in α reduces the overall weighted price, increasing welfare. If marginal cost has an intermediate slope, welfare initially decreases for low α but increases once real-time prices have sufficiently converged for high α.

The ambiguous change of RTP consumers' welfare with respect to α is in slight contrast with the result of Borenstein and Holland (2005).[9] The difference does not distort consequent results, as the consumer welfare of fixed pricing consumers always increases by relatively more — even in cases in which RTP consumers also gain. As a result, the welfare gain of switchers remains strictly positive and diminishes in α as stated in part (iii). As α increases, real-time prices converge, reducing the relative arbitrage opportunity of RTP. By contrast, higher substitutability of electricity, captured by a larger σ, enhances consumers' ability to shift consumption across periods, thereby increasing the switching gains from RTP.

3.2 Endogenous RTP adoption

So far, we have treated α and thus retail contract adoption as exogenous. The second extension of this paper is to investigate the adoption rate of RTP on basis of consumers' incentive to switch, i.e., the welfare gain of switching introduced above. To do so, assume that each consumer n chooses their contract endogenously, but that adopting a real-time pricing contract comes with adoption cost c. This can represent a wide array of

[9]The difference is driven by the inclusion of capacity investment decisions by wholesale producers. Producers invest in capacity in the long-run, which allows them to reduce their marginal cost.

costs associated with RTP contracts, namely metering costs or alternative information or transaction costs that come with real-time electricity consumption. Assume each consumer's individual cost c_n to be of linear form

$$c_n = c_0 + \theta_i c_1, \tag{14}$$

where c_0, c_1 are constants and θ_i is continuously distributed among consumers with positive density in the interval $[0, \bar{\theta}]$. Hence, each consumer faces the same cost c_0, which represents uniform metering costs for all consumers. However, consumers are differently exposed to the cost term c_1, which represents cost factors that may be heterogeneous between consumers, such as information costs, risk aversion or other individual characteristics. For example, consumers, who are early adopters and well-informed about RTP and its risks, have a lower adoption cost than consumers, who do not have the technical know-how to inform themselves about real-time prices or who may be afraid of being exposed to volatile prices. The parameter θ hence captures the degree of affinity towards RTP contracts and introduces consumer heterogeneity in a reduced form.

A consumer adopts an RTP contract as long as the switching gain is larger than or equal to the consumer's individual cost c_n. Once again, we assume that each consumer constitutes a very small share of total demand, so that it does not take into account potential price differences due to its change in consumption. Moreover, consumers observe their own adoption costs, as well as the switching gain given current prices but not α. Then, the endogenous share of RTP consumers or *adoption rate*, denoted α^*, is pinpointed by

$$\alpha^* = P[c \leq \Delta\tilde{U}^S(\alpha^*, \sigma)], \tag{15}$$

where the right-hand side denotes the share of consumers with adoption costs below the existing switching gain. If α^* is larger than the right-hand side, some consumers on RTP have an adoption cost larger than the current switching gain. These consumers would switch back to a fixed pricing contract. Similarly, if α^* is lower, there are some consumers with an adoption cost lower than the current switching gain, who are better off adopting RTP. Therefore, only when both shares are equal, no consumer deviates.

Lastly, consider the corner cases. If the maximum cost is lower than the minimum switching gain, $c_0 + \bar{\theta}c_1 \leq \min(\Delta\tilde{U}^S)$, all consumers switch and $\alpha^* = 1$. If the welfare gain is smaller than the baseline adoption cost c_0, all consumers remain on fixed pricing contracts and $\alpha^* = 0$. Proposition 3 summarizes the results in a setting with competitive retail markets (no retail market frictions).

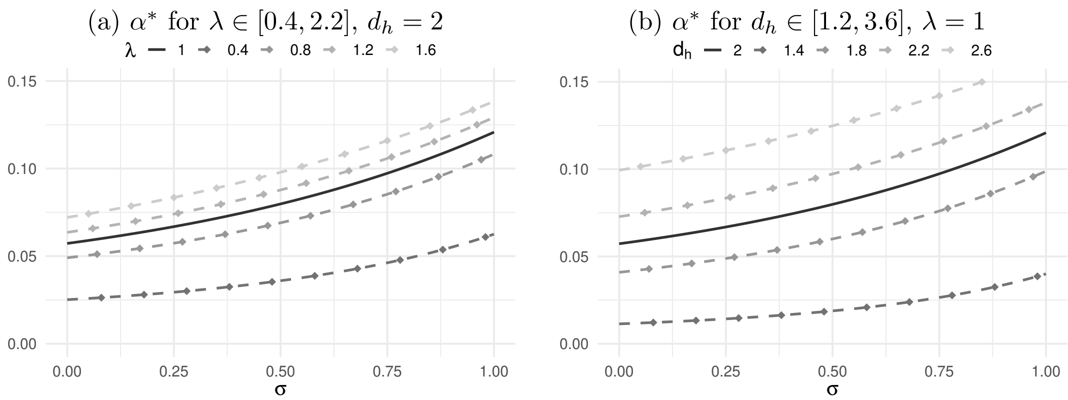
Proposition 3. *Given perfectly competitive retailers and heterogeneous adoption cost, there exists a unique equilibrium with endogenous adoption rate of real-time pricing contracts, $\alpha_p^* \in [0, 1]$. The adoption rate α_p^* increases in the substitutability of electricity.*

Proof. Proposition 3 follows from Proposition 2. Since $\Delta\tilde{U}_p^S$ is strictly positive and decreasing in α, there is a unique endogenous adoption rate α_p^*. Lastly, since $\Delta\tilde{U}_p^S$ increases in σ, the equality in (15) only holds if α^* increases as well. Thus, the endogenous adoption rate of RTP contracts increases with the substitutability of electricity. $\square$

Proposition 3 highlights the first factor that restricts the adoption of RTP contracts: the lack of flexibility in consumption. When consumers cannot effectively substitute consumption between different time periods, they only react to price variations in a limited fashion. Hence, the benefits of adopting an RTP contract are minimal. As there are adoption costs associated with these contracts, this results in a lower adoption rate.

Figure 1 shows the equilibrium adoption rate for different parameter values. As the graph is based on stylized parameter values, our attention is on the trend of the curves and their relative differences rather than their absolute values. In both panels, the solid line depicts the benchmark. As Proposition 3 suggests, the line is upward-sloping, indicating that the adoption rate increases in the substitutability of electricity, which is depicted on the x-axis. Moreover, panel (a) shows the adoption rate for different λ, the slope of the marginal cost curve, and panel (b) for different d_h, consumers' preference for electricity consumption in the peak period h. Panel (a) shows that an increase in λ leads to an upward shift in the adoption rate α^*. The slope of the marginal cost affects the welfare gain of switching $\Delta\tilde{U}_p^S$ through the real-time price difference. As λ rises, the supply curve becomes steeper, and with the demand function unchanged, the price difference increases. This boosts the switching gain and consequently the adoption rate. A similar mechanism drives the effect in panel (b). An increase in d_h shifts the peak demand curve outward, leading to greater price disparities with the supply function unchanged, which raises the switching gain and hence the adoption rate.

Figure 1: Equilibrium adoption rate for alternative model parameters



Note: The graphs are created with the following parameter values: $a = 1$, $d_l = 1$. Moreover, the adoption cost parameter θ is uniformly distributed between $[0, a]$ with $c_0 = 0$ and $c_1 = 1$.

3.3 Retail market frictions

The subsequent analysis introduces retail market frictions and investigates how they affect the adoption decision of RTP contracts. Specifically, we consider two different sources of distortions. First, we consider lump-sum grid fees (or taxes) that retailers pass on to final consumers. Second, we introduce differentiated retailers that have some degree of market power and thus the ability to charge markups to consumers.

Grid fees. Grid fees are collected by transmission and distribution grid owners to finance the operation and expansion of the power grid. In most electricity markets, these cost components are heavily regulated and paid by the final consumers. While paid by consumers, grid owners typically charge the fees to retailers, who then pass it on to their respective customers. Let us therefore denote the per-quantity grid fee as τ.[10] Then the retailers' objective function becomes

$$\pi^r = \alpha \sum_t (p_t - w_t - \tau) q_t(p_t, p_s) + (1-\alpha) \sum_t (\bar{p} - w_t - \tau) q_t(\bar{p}). \tag{16}$$

Retailers still compete in prices, which yields the following retail price function

$$\begin{aligned} p_t &= w_t + \tau, \quad \forall t, \\ \bar{p} &= \frac{w_l q_l(\bar{p}) + w_h q_h(\bar{p})}{q_l(\bar{p}) + q_h(\bar{p})} + \tau, \end{aligned} \tag{17}$$

which are equivalent to the original equilibrium prices (8) except for the additional uniform markup to cover grid fees.[11] The additional fee implies that consumers face higher retail prices than in the benchmark of perfect competition. This reduces demand and hence wholesale prices. While the general structure of the switching gain in (13) remains unchanged, the introduction of the uniform fee generates equilibrium effects that eventually reduce the disparity of real-time prices; bringing us to Proposition 4.

Proposition 4. *Given perfectly competitive retailers and heterogeneous adoption cost, an increase in grid fees τ decreases the adoption rate of real-time pricing contracts α^*.*

Proof. Proposition 4 follows from the observation that the real-time price difference (A7) is decreasing in the grid fee parameter τ. A reduction in the real-time price difference lowers the welfare gain from switching $\tilde{U}_p^S$, which in turn reduces the adoption rate α_p^*. □

Proposition 4 follows from demand in the peak period being more elastic (Assumption U1). With the introduction of a lump-sum fee, total demand in the peak period

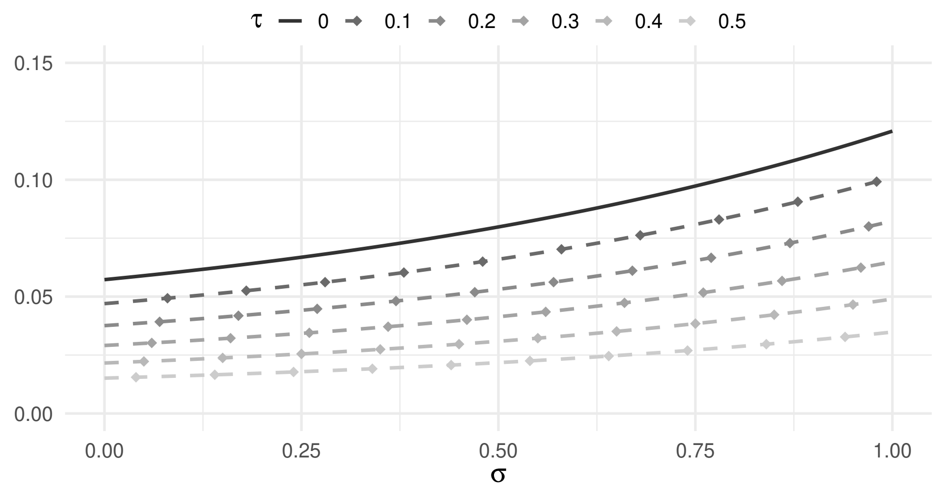
[10] τ can also be interpreted as a per-unit tax, which are typically part of consumers' electricity bill as well.

[11] The explicit expressions of the wholesale prices in terms of model parameters are given in Appendix A.2.

decreases more significantly than in the off-peak period. Consequently, the peak retail price increases less than the off-peak retail price, reducing the difference in real-time prices. As the switching gain is determined by the disparity of real-time prices, the partial convergence of real-time prices makes RTP contracts less appealing, leading to a lower adoption rate.

Figure 2 illustrates the result of Proposition 4. An increase in grid fees τ results in a downward shift of the adoption rate. Moreover, the adoption rate increases less in the substitutability of electricity when high grid fees exists. Since substitutability raises the benefits of RTP when prices are highly dispersed, a reduction in the real-time price difference also decreases the positive effect of substitutability on adoption rate. Our current argumentation is simplified due to the linearity of supply. If the marginal cost of production was convex, then the changes in retail prices are jointly determined by the respective demand and supply slopes. In Section 3.4, we introduce convex marginal costs and find that results remain robust.

Figure 2: Equilibrium adoption rate with grid fees



Note: The graph is created with the following parameter values: $a = 1$, $d_l = 1$, $d_h = 2$, $\lambda = 1$. Moreover, the adoption cost parameter θ is uniformly distributed between $[0, a]$ with $c_0 = 0$ and $c_1 = 1$.

Market power. In this section, we introduce differentiated retailers with market power. Market power in the retail market can be driven by numerous factors, such as switching costs, branding and behavioral patterns of consumers. For example, experimental evidence shows that many consumers do not switch contracts and remain with their default contractor despite considerable savings potential (e.g., Brennan, 2007; Gravert, 2024).

To introduce differentiated retailers with market power, we expand on Poletti and Wright (2020) and assume a standard Hotelling model, where consumers are uniformly distributed along a unit interval with two retailers (i, j) located at the endpoints of the interval. Consumers face linear transportation costs, represented by γ, which reflect the sources of market power mentioned above. The greater the value of γ, the more differentiated the retailers are, and the higher their market power. Both retailers provide fixed pricing and RTP contracts, leading them to compete on two distinct Hotelling lines. Assuming the market to be covered, the consumer indifferent between the retailers is given by

$$
\begin{aligned}
\tilde{U}(p_l^i, p_h^i) - \gamma\beta^R &= \tilde{U}(p_l^j, p_h^j) - \gamma(1-\beta^R), \\
\tilde{U}(\bar{p}^i) - \gamma\beta^F &= \tilde{U}(\bar{p}^j) - \gamma(1-\beta^F),
\end{aligned}
\tag{18}
$$

where β^R and β^F denote the indifferent consumer for the respective contracts. Rearranging (18) allows us to compute the market shares of retailers for both contracts. The market share of retailer i for both contracts is given by

$$
\begin{aligned}
\beta_i^R &= \frac{1}{2} + \frac{\tilde{U}(p_l^i, p_h^i) - \tilde{U}(p_l^j, p_h^j)}{2\gamma}, \\
\beta_i^F &= \frac{1}{2} + \frac{\tilde{U}(\bar{p}^i) - \tilde{U}(\bar{p}^j)}{2\gamma}.
\end{aligned}
$$

Retailers i's profits are then

$$
\begin{aligned}
\pi_i &= \alpha\beta_i^R\pi^R(p_l, p_h) + (1-\alpha)\beta_i^F\pi^F(\bar{p}), \\
\pi^R &= (p_l - w_l)q_l(p_l, p_h) + (p_h - w_h)q_h(p_h, p_l), \\
\pi^F &= (\bar{p} - w_l)q_l(\bar{p}) + (\bar{p} - w_h)q_h(\bar{p}),
\end{aligned}
\tag{19}
$$

where π^R and π^F are the per-consumer profits from RTP and fixed pricing contracts. Retailers treat input (wholesale) prices as given. We justify this assumption by considering a multitude of small retailers, for example municipal utilities, that do not have any market power in the wholesale market. Moreover, we assume that retailers do not take into account any changes in the share of RTP consumers when setting prices, i.e., they do not actively induce switching between contract types through their pricing. This implies that retailers are indifferent between having a fixed pricing or RTP consumer. While simplifying, this assumption is in line with the underlying model structure. Should retailers have a preference for one of the two contract types, they would only offer the preferred one. Since we do not model the contract choices of retailers and assume that all retailers offer both contract types, retailers only consider switching between retailers (β) but not between contract types (α).

Differentiating (19) with respect to prices, applying symmetry and using $\partial\tilde{U}/\partial p_t = -q_t$ for all t as well as $\partial\tilde{U}/\partial\bar{p} = -(q_l + q_h)$, yields the first-order conditions of retailers

$$
\begin{aligned}
\frac{\partial \pi^R}{\partial p_l} &= \frac{1}{\gamma} q_l(p_l, p_h)\pi^R(p_l, p_h), \\
\frac{\partial \pi^R}{\partial p_h} &= \frac{1}{\gamma} q_h(p_h, p_l)\pi^R(p_l, p_h), \\
\frac{\partial \pi^F}{\partial \bar{p}} &= \frac{1}{\gamma} [q_l(\bar{p}) + q_h(\bar{p})]\pi^F(\bar{p}).
\end{aligned} \tag{20}
$$

The first-order conditions have an intuitive interpretation. Retailers set prices such that two effects offset each other. The left-hand side represents the increase in the per-consumer profits as a retailer increases prices. The right-hand shows the loss in profits, as consumers switch to the other retailer due to increase in prices, which is driven by the differentiation parameter γ. Retailer profits are maximized if both effects cancel each other out.

Let us provide some additional intuition by considering the limit cases. If $\gamma = 0$, there is no 'transportation cost' and retailers are not differentiated. Then, all consumers would immediately switch to the other retailer as a response to price increases. Hence, consumers simply buy from the retailer with the lowest prices and we return to the original equilibrium conditions (8), where retailers set prices equal to wholesale prices — the standard Bertrand framework. If γ approaches ∞, transportation costs are so high that consumers do not switch due to price increases. Hence, retailers set prices such that their per-consumer profits π^R and π^F are maximized. This coincides with the profit-maximizing conditions of a retail monopolist, which can be found in Appendix B.1. Intuitively, if retailers are fully differentiated, both retailers serve two separate markets, where both have monopolistic market power.

In the limit $\gamma \to \infty$, the retail prices that maximize profits implied by (20) are

$$
\begin{aligned}
p_t(w_t) &= \frac{a + w_t}{2}, \quad \forall t, \\
\bar{p}(w_l, w_h) &= \frac{2d_l^2(a + w_l) + 2d_h^2(a + w_h) - \sigma d_l d_h(2a + w_l + w_h)}{4(d_l^2 + d_h^2 - \sigma d_l d_h)}
\end{aligned} \tag{21}
$$

which are identical to the typical monopoly solution.[12]

Similar to grid fees, market power in the retail market drives a wedge between retail and wholesale prices and leads to markups. By plugging in monopolistic retail prices (21) into the switching gain in (12), we obtain the welfare gain of switchers under retail *monopoly*

$$
\Delta \tilde{U}_m^S(p_l, p_h) = \frac{d_l^2 d_h^2 (w_h - w_l)^2}{16(d_l^2 + d_h^2 - d_l d_h \sigma)} = \frac{d_l^2 d_h^2 (p_h - p_l)^2}{4(d_l^2 + d_h^2 - d_l d_h \sigma)}, \tag{22}
$$

[12]The derivations of the prices can be found in Appendix B.1 and the explicit expression of wholesale prices in Appendix A.3.

which is positive and has the same general form as the switching gain under perfect competition. The gain of switchers is still determined by the difference in real-time prices, only that retail prices are now set by a profit-maximizing retail monopoly. Further, we derive the adoption rate of RTP contracts with a retail monopoly, denoted α_m^*.

Proposition 5. *Given a retail monopoly* ($\gamma \to \infty$) *and heterogeneous adoption cost, there exists an equilibrium with endogenous adoption rate of real-time pricing contracts,* $\alpha_m^* \in (0, 1]$. *The adoption rate* α_m^* *increases in the substitutability of electricity. The adoption rate is lower than under perfect competition,* $\alpha_m^* < \alpha_p^*$.

Proof. See proof in appendix. □

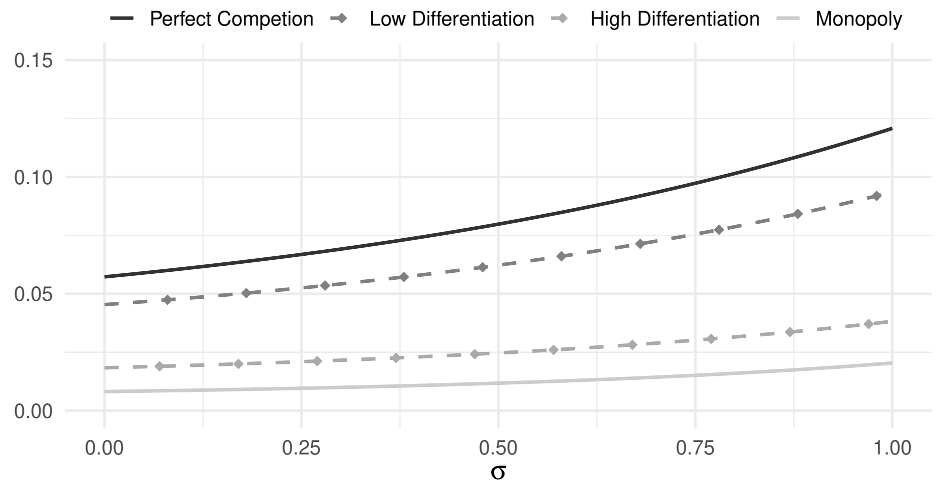
Proposition 5 shows that most characteristics of the equilibrium with perfect competition remain in a retail monopoly. In the proof, we show that $\Delta\tilde{U}_m^S$ has the same properties as $\Delta\tilde{U}_p^S$ except for being strictly lower. In other words, market power among retailers lowers the attractiveness of RTP contracts.

This is driven by two distinct mechanisms. First, since peak demand is more elastic, introducing a markup reduces peak demand relatively stronger than off-peak demand. This reduces the difference in real-time wholesale prices and is identical to the lump-sum grid fee scenario before. Secondly and unlike before, the markup charged by a retail monopolist is not uniform in every time period. Since off-peak demand is less elastic than peak demand, the markup charged on off-peak prices is relatively higher than on peak prices.[13] This leads to an additional reduction in the difference of real-time retail prices. Recall that the benefits of RTP contracts —and the gains of flexible consumption— are especially large if high price disparities exist. By partially muting any real-time price differences with considerable markups, the adoption rate of RTP contracts drops under a retail monopoly.

Figure 3 shows the impact of market power in the retail market on the adoption rate. The solid black line indicates the adoption rate in a perfectly competitive retail market, while the solid grey line shows the adoption rate under a retail monopoly. The dashed lines show the adoption rate for varying degrees of differentiation/market power. The adoption rate shifts downward as market power of retailers increases and is lowest when retailers have monopolistic pricing power. While all lines slope upwards, indicating that substitutability increases the adoption rate, the slope is more pronounced when retailers are perfectly competitive. The intuition is the same as with grid fees. Differentiated retailers charge markups which mute real-time price differences and thus the additional benefits of flexibility.

[13] This becomes evident when subtracting wholesale prices from equation (21).

Figure 3: Equilibrium adoption rate with retailer market power



Note: The graph is created with the following parameter values: $a = 1$, $d_l = 1$, $d_h = 2$, $\lambda = 1$ and $\gamma = 0.1$ for low differentiation and $\gamma = 0.5$ for high differentiation. Moreover, the adoption cost parameter θ is uniformly distributed between $[0, a]$ with $c_0 = 0$ and $c_1 = 1$.

3.4 Robustness

This section illustrates the robustness of the results by computing equilibrium outcomes for alternative specifications. Specifically, we discuss alternative production cost and adoption cost functions.

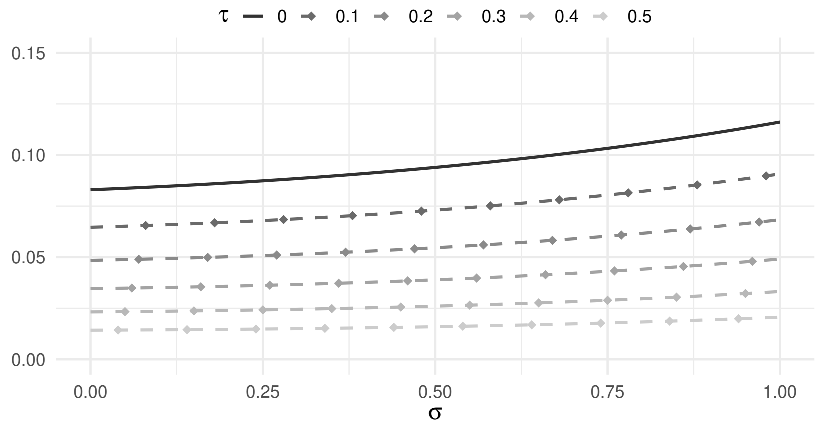
Convex marginal cost. A helpful assumption in our model setup to obtain analytical results is the linearity of marginal cost. However, a better representation of marginal cost curves in electricity markets may be a convex marginal cost function. This section represents the results for quadratic marginal costs.

Figure 4 presents the adoption rate given retail market frictions when marginal cost is increasing quadratically. Panel (a) of Figure 4 shows the adoption rate with lump-sum fees, while panel (b) depicts it with retailer market power. The solid black line represents the benchmark scenario with no retail market frictions, meaning neither distribution costs nor market power. The plots show that the main results of Proposition 3-5 remain unchanged. The adoption rate continues to rise with the substitutability of electricity. Additionally, any retail market friction that creates a gap between retail and wholesale prices diminishes the adoption rate.

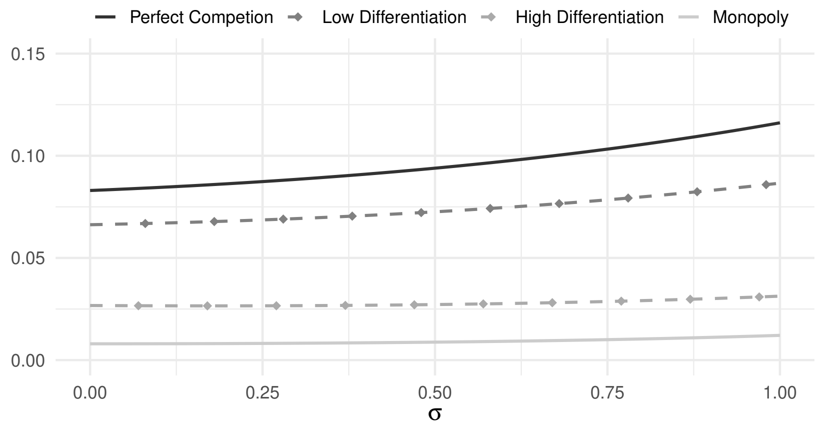
The intuition is similar to before. In the baseline with linear marginal cost, the decrease in the adoption rate was due to peak demand being more elastic, causing the peak retail price to increase by less than the off-peak retail price when a markup is charged in the retail market. With convex marginal cost, the supply curve is steeper in the peak period compared to the off-peak period. Consequently, a retail markup leads to an additional reduction in the wholesale peak price relative to the off-peak price. Thus, the real-time price difference decreases faster with convex marginal cost than in the linear case. In summary, the main findings remain robust or are even more pronounced with convex marginal cost.

Figure 4: Equilibrium adoption rate with quadratic marginal cost

(a) Grid fees



(b) Market power

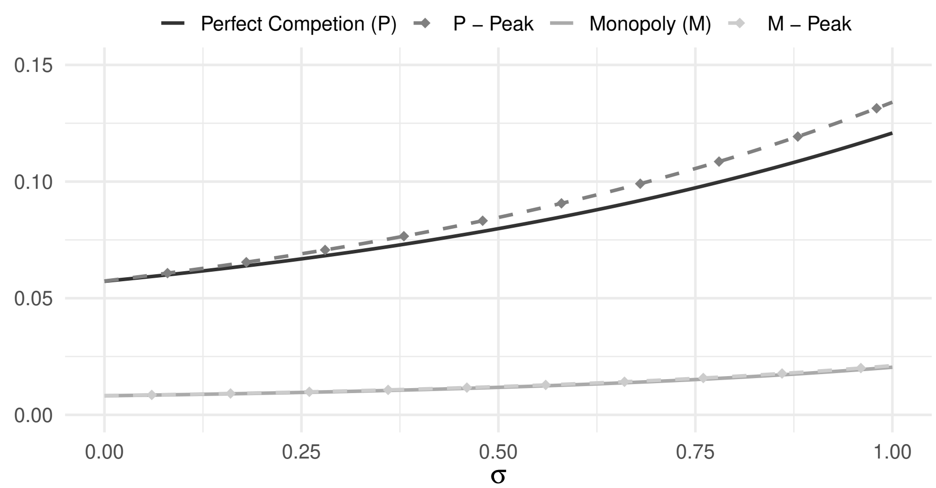


Note: The graph is created with the following parameter values: $a = 1$, $d_l = 1$, $d_h = 2$. Marginal cost is given by $C_q(q) = q^\lambda$ with $\lambda = 2$. Moreover, the adoption cost parameter θ is uniformly distributed between $[0, a]$ with $c_0 = 0$ and $c_1 = 1$.

Disutility from high peak prices. Our adoption cost specification is a reduced-form method to capture the varying degrees of aversion consumers may have towards RTP. One reason, so far not discussed, is risk aversion (Boom and Schwenen, 2021). These consumers prefer to avoid exposure to volatile and potentially steep prices. Since our model is fully deterministic, we cannot explicitly account for risk aversion without major adjustments. However, we offer some illustrative evidence demonstrating how risk aversion might interact with flexibility of consumption. If risk aversion serves as a major barrier to RTP, consumers dislike price volatility; especially the risk of encountering inevitable high peak prices. To reflect this in our consumer-specific adoption cost, we treat the term c_1 as a function dependent on the peak price, denoted $c_1(p_h)$. Therefore, the average adoption cost for RTP increases with high peak prices and decreases with low peak prices. Although this approach is not identical to modeling risk aversion, it assumes that consumers experience additional 'disutility' from high peak prices, resembling a partial effect of risk aversion.

Figure 5 illustrates the adoption rate when the adoption cost is dependent on peak prices. The solid lines represent the baseline under perfect competition and monopoly scenarios, while the dashed lines show the corresponding adoption rate when consumers have an aversion to high peak prices. The function $c_1(p_h)$ is calibrated such that the adoption rate matches the baseline when $\sigma = 0$, facilitating easier comparison. For perfectly competitive retailers, rising substitutability of electricity leads to a more pronounced increase in the adoption rate if consumers dislike extreme peak prices. With some consumers on RTP, an increase in σ shifts more consumption from peak to off-peak periods, reducing peak load and consequently peak prices. Therefore, an increase in substitutability not only increases the gains of RTP but also lowers the adoption cost, leading to a greater increase in the adoption rate when consumers dislike high peak prices. However, this does not fully materialize when retailers set monopoly prices. By imposing significant markups during off-peak periods, consumers are less inclined to shift their consumption to off-peak periods. Peak prices, thus, do not necessarily decrease further, and the outcomes are nearly identical to the baseline.

Figure 5: Equilibrium adoption rate with disutility from peak prices



Note: The graph is created with the following parameter values: $a = 1,\ d_l = 1,\ d_h = 2\ \lambda = 1$. Moreover, the adoption cost parameter θ is uniformly distributed between $[0, a]$ with $c_0 = 0$ and $c_1 = 1$ for the baseline cases. Otherwise, $c_1 = 1.36p_h$ for *P-Peak* and $c_1 = 1.29p_h$ for *M-Peak*, where the coefficients were chosen to ensure that α^* is identical with the baseline if $\sigma = 0$.

4 Conclusion

This paper examines why real-time pricing (RTP), despite its theoretical efficiency and policy relevance, remains scarcely adopted by residential consumers. By extending the stylized market model of Borenstein and Holland (2005) to include intertemporal demand flexibility, endogenous contract choice, and retail market frictions, we show that RTP adoption crucially depends on the interaction between consumer flexibility and retail price design. While higher flexibility increases the potential gains from RTP, these gains are strongly reduced when retail prices are distorted by lump-sum fees, taxes, or markups. Such frictions dampen real-time price volatility through equilibrium demand responses, thereby weakening the arbitrage opportunities that make RTP attractive.

These results carry important policy implications. Promoting flexibility through electrification or automation alone may not suffice to induce widespread RTP adoption if retail price structures mute time-varying signals. In particular, charging grid fees and taxes as lump sums reduces retail price volatility and limits the benefits of RTP, whereas time-varying charges can preserve these incentives. Similarly, a default retail contractor setup or perceived high switching costs can enable retailers to charge markups, which have similar muting effects. More generally, our findings suggest that retail market design is central to realizing the efficiency gains from RTP and to aligning consumer incentives with the needs of a renewable-based electricity system.

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A Equilibrium solutions

A.1 Explicit price functions (Perfect Competition)

The explicit price functions in perfect competition are derived by solving the system of equilibrium conditions (5), (6), (7) and (8). Assume marginal cost to be of the form $C_q(q) = \lambda q$ and normalize $d_l = 1$.

$$p_l(\alpha,\sigma) = w_l(\alpha,\sigma) = \frac{a\lambda(1+d_h^2-d_h\sigma)(2+d_h^2\alpha\lambda-d_h\sigma)}{b(\alpha,\sigma)} \tag{A1}$$

$$p_h(\alpha,\sigma) = w_h(\alpha,\sigma) = \frac{a\lambda(1+d_h^2-d_h\sigma)(2d_h^2+d_h^2\alpha\lambda-d_h\sigma)}{b(\alpha,\sigma)} \tag{A2}$$

$$\bar{p}(\alpha,\sigma) = p_h(\alpha,\sigma) - \frac{a\lambda(2-d_h\sigma)(d_h^2-1)}{b(\alpha,\sigma)} \qquad \text{, where} \tag{A3}$$

$$\begin{aligned} b(\alpha,\sigma) = {} & (1+d_h^2-d_h\sigma+d_h^2\alpha\lambda)(4-\sigma^2)+\lambda[(1+d_h^2-d_h\sigma)(2d_h^2+d_h^2\alpha\lambda-d_h\sigma) \\ & -(2-d_h\sigma)(d_h^2-1)]. \end{aligned} \tag{A4}$$

To simplify later derivations, Lemma A1 summarizes properties of $b(\alpha,\sigma)$.

Lemma A1. *The term $b(\alpha,\sigma)$ increases in the share of real-time consumers α, decreases in the substitutability of electricity σ, and is positive.*

Proof. The derivatives of $b(\alpha,\sigma)$ with respect to the parameters α and σ are given by

$$\begin{aligned} \frac{\partial b}{\partial \alpha} &= d_h^2\lambda(4-\sigma^2)+d_h^2\lambda^2(1+d_h^2-d_h^2\sigma) > 0 \\ \frac{\partial b}{\partial \sigma} &= -d_h(4-\sigma^2)-2\sigma(1+d_h^2-d_h\sigma+d_h^2\alpha\lambda)-d_h\lambda(2+2d_h^2+d_h^2\alpha\lambda-2d_h\sigma) < 0 \end{aligned}$$

where the inequalities follow from Assumptions U1-U2. Since $b(\alpha,\sigma)$ increases in α and decreases in σ, we prove the positivity of $b(\alpha,\sigma)$ by showing $b(0,2/d_h) > 0$. Then, the term becomes

$$b(0,2/d_h) = (d_h^2-1)(4-4/d_h^2)+2\lambda(d_h^2-1)^2$$

which is positive due to Assumption U1. $\square$

A.2 Explicit price functions (Grid fees)

The explicit price functions with grid fees are derived by solving the system of equilibrium conditions (5), (6), (7) and (17). Assume marginal cost to be of the form $C_q(q) = \lambda q$ and normalize $d_l = 1$.

$$w_l(\alpha,\sigma) = \frac{(a-\tau)\lambda(1+d_h^2-d_h\sigma)(2+d_h^2\alpha\lambda-d_h\sigma)}{b(\alpha,\sigma)} \tag{A5}$$

$$w_h(\alpha,\sigma) = \frac{(a-\tau)\lambda(1+d_h^2-d_h\sigma)(2d_h^2+d_h^2\alpha\lambda-d_h\sigma)}{b(\alpha,\sigma)} \tag{A6}$$

$$w_h(\alpha,\sigma) - w_l(\alpha,\sigma) = \frac{2(a-\tau)\lambda(1+d_h^2-d_h\sigma)(d_h^2-1)}{b(\alpha,\sigma)} \tag{A7}$$

A.3 Explicit price functions (Monopoly)

The explicit price functions in a retail monopoly are derived by solving the system of equilibrium conditions (5), (6), (7) and (21). Assume marginal cost to be of the form $C_q(q) = \lambda q$ and normalize $d_l = 1$.

$$w_l(\alpha,\sigma) = \frac{a\lambda(1 + d_h^2 - d_h\sigma)(4 + d_h^2\alpha\lambda - 2d_h\sigma)}{b_m(\alpha,\sigma,d_h)} \tag{A8}$$

$$w_h(\alpha,\sigma) = \frac{a\lambda(1 + d_h^2 - d_h\sigma)(4d_h^2 + d_h^2\alpha\lambda - 2d_h\sigma)}{b_m(\alpha,\sigma,d_h)} \quad \text{, where} \tag{A9}$$

$$\begin{aligned} b_m(\alpha,\sigma) = {} & 2(2 + 2d_h^2 - 2d_h\sigma + d_h^2\alpha\lambda)(4 - \sigma^2) + \lambda[(1 + d_h^2 - d_h\sigma)(4d_h^2 + d_h^2\alpha\lambda - 2d_h\sigma) \\ & - 2(2 - d_h\sigma)(d_h^2 - 1)]. \end{aligned} \tag{A10}$$

To simplify later derivations, Lemma A2 summarizes properties of $b_m(\alpha,\sigma)$.

Lemma A2. *The term $b_m(\alpha,\sigma)$ increases in the share of real-time consumers α, decreases in the substitutability of electricity σ, and is positive.*

Proof. The derivatives of $b_m(\alpha,\sigma)$ with respect to the parameters α and σ are given by

$$\frac{\partial b_m}{\partial \alpha} = 2d_h^2\lambda(4 - \sigma^2) + d_h^2\lambda^2(1 + d_h^2 - d_h^2\sigma) > 0$$

$$\frac{\partial b_m}{\partial \sigma} = -4d_h(4 - \sigma^2) - 4\sigma(2 + 2d_h^2 - 2d_h\sigma + d_h^2\alpha\lambda) - d_h\lambda(4 + 4d_h^2 + d_h^2\alpha\lambda - 4d_h\sigma) < 0$$

where the inequalities follow from Assumptions U1-U2. Since $b_m(\alpha,\sigma)$ increases in α and decreases in σ, we prove the positivity of $b_m(\alpha,\sigma)$ by showing $b(0, 2/d_h) > 0$. Then, the term becomes

$$b_m(0, 2/d_h) = (d_h^2 - 1)(4 - 4/d_h^2) + 2\lambda(d_h^2 - 1)^2$$

which is positive due to Assumption U1. □

B Proofs

Proof of Proposition 1. We prove part (i)-(iv) separately.

Part (i). Part (i) is shown by taking the partial derivative of real-time prices, (A1) and (A2),

$$\frac{\partial p_l}{\partial \alpha} = \frac{ad_h^2\lambda^2(d_h^2-1)(1+d_h^2-d_h\sigma)(4+2d_h^2\lambda-d_h\sigma\lambda-\sigma^2)}{b(\alpha,\sigma)^2} > 0,$$

since $(4+2d_h^2\lambda-d_h\sigma\lambda-\sigma^2) > 0$ due to Assumptions U1-U2. Equivalently,

$$\frac{\partial p_h}{\partial \alpha} = \frac{ad_h^2\lambda^2(d_h^2-1)(1+d_h^2-d_h\sigma)(-4-2\lambda+d_h\sigma\lambda+\sigma^2)}{b(\alpha,\sigma)^2} < 0,$$

since $(-4-2\lambda+d_h\sigma\lambda+\sigma^2) < 0$ for all $\sigma \in [0,\bar{\sigma}]$.

Part (ii). Part (ii) is shown by taking the partial derivative of the fixed price, (A3),

$$\frac{\partial \bar{p}}{\partial \alpha} = \frac{ad_h^2\lambda^2(d_h^2-1)^2(\sigma^2-4)}{b(\alpha,\sigma)^2} < 0,$$

due to Assumptions U1-U2.

Part (iii). We show $\frac{\partial^2(p_h-p_l)}{\partial\alpha\partial\sigma} < 0$. The first-order partial derivative of the price difference is

$$\begin{aligned}\frac{\partial(p_h-p_l)}{\partial\sigma} &= \frac{\partial p_h}{\partial\sigma} - \frac{\partial p_l}{\partial\sigma} \\ &= \frac{2a\lambda(d_h^2-1)\{2\sigma(1+d_h^2-d_h\sigma)^2+(1-\alpha)\lambda[4d_h^3-2d_h^2\sigma(1+d_h^2-d_h\sigma)]\}}{b(\alpha,\sigma)^2},\end{aligned} \tag{B1}$$

which is positive since all terms are positive. Note that the last term in the brackets is minimized at $\sigma = 1/d_h + 2d_h$, which is greater than 2 and thus greater than $\bar{\sigma}$. Since the term is positive at the bounds of $\sigma \in [0,\bar{\sigma}]$, it is positive for all σ.

Hence, the numerator of (B1) decreases in α. From Lemma A1, $b(\alpha,\sigma)$ increases in α. Thus, $\frac{\partial(p_h-p_l)}{\partial\sigma}$ decreases in α, which implies $\frac{\partial^2(p_h-p_l)}{\partial\sigma\partial\alpha} = \frac{\partial^2(p_h-p_l)}{\partial\alpha\partial\sigma} < 0$.

Part (iv). We show $\frac{\partial^2\bar{p}}{\partial\alpha\partial\sigma} < 0$. The second-order partial derivative of the fixed price, (A3), is

$$\frac{\partial^2\bar{p}}{\partial\alpha\partial\sigma} = \frac{2ad_h^2\lambda^2(d_h^2-1)^2x(\alpha,\sigma)}{b(\alpha,\sigma)^3},$$

$$x(\alpha,\sigma) = y(\sigma) + \lambda z(\alpha,\sigma),$$

$$y(\sigma) = -2d_h\sigma^4 + (1+d_h^2)\sigma^3 + 12d_h\sigma^2 - (4d_h^2+4)\sigma - 16d_h, \tag{B2}$$

$$z(\alpha,\sigma) = (d_h^4\sigma - 4d_h^3)(2+\alpha\lambda) + \alpha(\sigma^2-4+d_h^2\sigma\lambda) - d_h^2\sigma^3 + 8d_h^2\sigma + 2\sigma - 8d_h. \tag{B3}$$

Following Lemma A1, $b(\alpha,\sigma)^3$ is positive. To show $x(\alpha,\sigma) < 0$, we proceed in two steps.
Step 1: We show $z(\alpha,\sigma) < 0$. First, differentiating (B3) w.r.t α yields

$$\frac{\partial z(\alpha,\sigma)}{\partial \alpha} = (\sigma^2 - 4) + \lambda(d_h^4\sigma - 4d_h^3 + d_h^2\sigma),$$

which is negative due to Assumption U2. It is thus sufficient to show $z(0,\sigma) < 0$. Setting $\alpha = 0$, (B3) becomes

$$z(0,\sigma) = -d_h^2\sigma^3 + 2d_h^4\sigma + 8d_h^2\sigma + 2\sigma - 8d_h^3 - 8d_h, \tag{B4}$$

with extreme points at

$$\sigma_{1,2} = \pm\sqrt{\frac{2}{3}\left(d_h^2 + \frac{1}{d_h^2}\right) + \frac{8}{3}}.$$

Both extreme points are out of bounds, $\sigma \in [0,\bar{\sigma}]$. To see $|\sigma_{1,2}| > \bar{\sigma} = 2/d_h$, square the inequality and rearrange

$$\frac{2}{3}(d_h^4 + 1) + \frac{8}{3} > 4,$$

which holds due Assumption U1. Since $z(\alpha,\sigma)$ is monotone within the interval $\sigma \in [0,\bar{\sigma}]$, it is sufficient to show that the inequality holds at the bounds. For $\sigma = 0$, equation (B4) is clearly negative. For $\sigma = 2/d_h$, the equation simplifies to $2d_h^2 - d_h^4 - 1$, which is negative for $d_h > 1$. Thus, $z(\alpha,\sigma) < 0$.
Step 2: We show $y(\alpha,\sigma) < 0$. Note that (B2) can be written as

$$y(\sigma) = (2-\sigma)(2+\sigma)(2d_h\sigma^2 - (d_h^2+1)\sigma - 4d_h)$$

with the roots

$$\sigma_{1,2} = \pm 2 \textbf{ and } \sigma_3 = \frac{1 + d_h^2 \pm \sqrt{1 + 34d_h^2 + d_h^4}}{4d_h}.$$

All roots lie outside the parameter bounds $\sigma \in [0,\bar{\sigma}]$. Note that

$$\frac{1 + d_h^2 + \sqrt{1 + 34d_h^2 + d_h^4}}{4d_h} > \frac{2}{d_h} \Leftrightarrow 1 + d_h^2 + \sqrt{1 + 34d_h^2 + d_h^4} > 8$$

holds due to Assumption U1. Since all roots lie outside the parameter interval, it is sufficient to show that $y(\sigma)$ is negative at the bounds. For $\sigma = 0$, equation (B2) is clearly negative. For $\sigma = 2/d_h$, the equation simplifies to $2d_h^2 - d_h^4 - 1$, which is equivalent to the condition in Step 1 and negative for $d_h > 1$. With $y(\sigma) < 0$ and $z(\alpha,\sigma) < 0$, $x(\alpha,\sigma) < 0$. Therefore $\partial^2\bar{p}/\partial\alpha\partial\sigma < 0$. □

Proof of Proposition 2. We prove part (i)-(iv) separately.

Part (i): We show $\frac{d\tilde{U}^F}{d\alpha} > 0$. Taking the derivative of $\tilde{U}^F$ with respect to α gives $\frac{d\tilde{U}^F}{d\alpha} = \frac{d\tilde{U}^F}{d\bar{p}}\frac{d\bar{p}}{d\alpha}$. Since $\frac{d\tilde{U}^F}{d\bar{p}} < 0$ and $\frac{d\bar{p}}{d\alpha} < 0$ from Proposition 1, $\frac{d\tilde{U}^F}{d\alpha} > 0$.

Part (ii): We show $\frac{d\tilde{U}^{RTP}}{d\alpha} < 0$. In terms of parameters, using (A1)-(A2), the derivative becomes

$$\frac{d\tilde{U}^{RTP}}{d\alpha} = -\frac{2a^2\lambda^2(d_h^2-1)^2 f(\alpha,\sigma)}{b(\alpha,\sigma)^3}, \text{where}$$
$$f(\alpha,\sigma) = (\sigma^2-4)[1+d_h^2-d_h\sigma+\lambda d_h^2(\alpha-1)] + \lambda^2(d_h^4-d_h^3\sigma+d_h^2)$$

which is negative when $f(\alpha,\sigma)$ is positive. The first product of $f(\alpha,\sigma)$ is always negative due to Assumptions U1-U2. Similarly, the second term is always positive. Both terms increase in λ, thus $f(\alpha,\sigma)$ is only positive if the marginal cost slope λ is sufficiently large.

We compute the threshold $\tilde{\lambda}$ so that $\tilde{U}^{RTP}$ decreases in α for all α and σ if $\lambda > \tilde{\lambda}$. As $f(\alpha,\sigma)$ decreases in α, we compute the threshold if $\alpha = 1$. To compute $\tilde{\lambda}$, solve $f(1,\sigma) = 0$, which yields

$$\tilde{\lambda}^2 = \frac{(4-\sigma^2)(1+d_h^2-d_h\sigma)}{d_h^4-d_h^3\sigma+d_h^2}.$$

Since $f(\alpha,\sigma)$ increases in λ, $f(\alpha,\sigma)$ is positive and $\frac{d\tilde{U}^{RTP}}{d\alpha}$ negative if $\lambda > \tilde{\lambda}$. If $\lambda \le \tilde{\lambda}$, $f(\alpha,\sigma)$ may be positive or negative depending on α. As $f(\alpha,\sigma)$ decreases in α, it may be positive for low α and become positive for high α, i.e., $\tilde{U}^{RTP}$ first decreases and then increases in α. If λ is sufficiently low, $\tilde{U}^{RTP}$ increases for all α.

Part (iii) We show properties of $\Delta\tilde{U}_p^S$. Using price functions (A1)-(A2), $\Delta\tilde{U}_p^S$ can be expressed in terms of model parameters

$$\Delta\tilde{U}_p^S(\alpha,\sigma) = \frac{a^2\lambda^2 d_h^2(d_h^2-1)^2(1+d_h^2-d_h\sigma)}{b(\alpha,\sigma)^2}. \tag{B5}$$

All terms in in equation (B5) are positive due to Assumptions U1-U2 and Lemma A1. Additionally, Lemma A1 shows that $b(\alpha,\sigma)$ increases in α. Hence, b^2 increases in α as well and $\Delta\tilde{U}_p^S(\alpha,\sigma)$ decreases in α.

Lastly, we want to show $\frac{d\Delta\tilde{U}_p^S}{d\sigma} > 0$. The derivative with respect to σ is given by

$$\frac{d\Delta\tilde{U}_p^S}{d\sigma} = \frac{a\lambda d_h^2(d_h^2-1)^2 l(\alpha,\sigma)}{b(\alpha,\sigma)^3}, \text{ with} \tag{B6}$$
$$l(\alpha,\sigma) = m(\sigma) + d_h\lambda n(\alpha,\sigma), \tag{B7}$$
$$m(\sigma) = 4d_h(d_h^2+1) + \sigma(5d_h^2\sigma^2 - 9\sigma(d_h^3+d_h) + 4(d_h^4+d_h^2+1), \tag{B8}$$
$$\begin{aligned} n(\alpha,\sigma) = {} & 2 + (8-4\alpha)d_h^2 + 2d_h^4 - (6-4\alpha)d_h\sigma(d_h+1) \\ & + (1-\alpha)3d_h^2\sigma^2 + d_h\alpha\lambda(d_h^3+d_h-d_h\sigma) \end{aligned} \tag{B9}$$

Step 1: We show $m(\sigma) > 0$ for all $\sigma \in [0, \bar{\sigma}]$. The roots of $m(\sigma)$ are given by

$$\sigma_1 = \frac{1 + d_h^2}{d_h} \textbf{ and } \sigma_{2,3} = 2\frac{1 + d_h^2 \pm \sqrt{1 + 7d_h^2 + d_h^4}}{5d_h}.$$

Both σ_1 and σ_2 are greater than 2 following Assumption $U1$, $d_h > 1$, and thus out of bounds. The root σ_3 is always negative. Since $m(0)$ is positive, $m(\sigma)$ is positive for all $\sigma \in [0, \bar{\sigma}]$.

Step 2: We show $n(\sigma, \alpha) > 0$ for all $\sigma \in [0, \bar{\sigma}]$. Since $n(\alpha, \sigma)$ is linear in α, it is sufficient to show positivity at the bounds of α. For $\alpha = 0$,

$$n(0, \sigma) = 2 + 8d_h^2 + 2d_h^4 - 6d_h\sigma(d_h + 1) + 3d_h^2\sigma^2,$$

with roots

$$\sigma_{1,2} = 2\frac{3d_h + 3d_h^2 \pm \sqrt{3}\sqrt{d_h^2 + 6d_h^3 - 5d_h^4 - 2d_h^6}}{3d_h^2},$$

which are not well-defined for $d_h > 1$. Thus, $n(0, \sigma)$ is positive for all σ. Similarly, for $\alpha = 1$ the expression becomes

$$n(1, \sigma) = 2 + 4d_h^2 + 2d_h^4 - 2d_h\sigma(d_h + 1) + \lambda(d_h^3 + d_h - d_h\sigma),$$

which is strictly decreasing in σ. Plugging in the upper limit $\bar{\sigma} = 2/d_h$,

$$n(1, 2/d_h) = 2 + 4d_h^2 + 2d_h^4 - 4(d_h + 1) + d_h\lambda(d_h^3 + d_h - 2/d_h),$$

which is positive due to Assumption U1. Thus, $n(1, \sigma)$ is positive for all σ. Hence, $n(\alpha, \sigma) > 0$. With $m(\sigma) > 0$ and $n(\alpha, \sigma) > 0$, $l(\alpha, \sigma) > 0$. Therefore $\frac{d\Delta\tilde{U}_p^S}{d\sigma} > 0$. □

Proof of Proposition 5. We show properties of $\Delta\tilde{U}_m^S$. Using price functions (A8)-(A9), $\Delta\tilde{U}_m^S$ can be expressed in terms of model parameters

$$\Delta\tilde{U}_m^S(\alpha,\sigma) = \frac{a^2\lambda^2 d_h^2(d_h^2-1)^2(1+d_h^2-d_h\sigma)}{b_m(\alpha,\sigma)^2}, \tag{B10}$$

which is identical to the consumer welfare in perfect competition (B5) except for the denominator, which is now denoted by b_m. From Lemma *A2*, we know that b_m is positive and increasing in α. Hence, consumer welfare decreases in α, which proves existence and uniqueness. Moreover, b_m is strictly greater than b, which implies that $\Delta\tilde{U}_m^S < \Delta\tilde{U}_p^S$. Thus, $\alpha_m^* < \alpha_p^*$.

To verify that $\tilde{U}_m^S$ increases in σ, it is sufficient to show that the price difference increases in σ. The partial derivative of the wholesale real- time prices (A8)-(A9) is

$$\frac{\partial w_h}{\partial\sigma} - \frac{\partial w_l}{\partial\sigma} = \frac{8a\lambda(d_h^2-1)\{4\sigma(1+d_h^2-d_h\sigma)^2+(1-\alpha)\lambda[4d_h^3-2d_h^2\sigma(1+d_h^2-d_h\sigma)]\}}{b_m(\alpha,\sigma)^2} > 0,$$

which holds due to Assumptions U1-U2. □

B.1 Profit maximisation of retail monopoly

A retail monopoly maximises sector profits (2)

$$\max_{p_l,p_h,\bar{p}} \alpha[(p_l - w_l)q_l(p_l, p_h) + (p_h - w_h)q_h(p_h, p_l)] + (1-\alpha)[(\bar{p} - w_l)q_l(\bar{p}) + (\bar{p} - w_h)q_h(\bar{p})].$$

The first-order conditions are given by

$$\begin{aligned}
\frac{\partial \pi^r}{\partial p_l} &= \alpha \left[q_l(p_l, p_h) + (p_l - w_l)\frac{\partial q_l}{\partial p_l} + (p_h - w_h)\frac{\partial q_h}{\partial p_l} \right] = 0, \\
\frac{\partial \pi^r}{\partial p_h} &= \alpha \left[q_h(p_h, p_l) + (p_h - w_h)\frac{\partial q_h}{\partial p_h} + (p_l - w_l)\frac{\partial q_l}{\partial p_h} \right] = 0, \\
\frac{\partial \pi^r}{\partial \bar{p}} &= (1-\alpha) \left[q_l(\bar{p}) + q_h(\bar{p}) + (\bar{p} - w_l)\frac{\partial q_l}{\partial \bar{p}} + (\bar{p} - w_h)\frac{\partial q_h}{\partial \bar{p}} \right] = 0.
\end{aligned}$$

Plugging in demand function (5) and rearranging

$$p_l = \frac{a + w_l}{2} - \frac{\sigma}{4}\frac{d_h}{d_l}(a + w_h - 2p_h), \tag{B11}$$

$$p_h = \frac{a + w_h}{2} - \frac{\sigma}{4}\frac{d_l}{d_h}(a + w_l - 2p_l), \tag{B12}$$

$$\bar{p} = \frac{2d_l^2(a + w_l) + 2d_h^2(a + w_h) - \sigma d_l d_h(2a + w_l + w_h)}{4(d_l^2 + d_h^2 - \sigma d_l d_h)}, \tag{B13}$$

where (B13) is already the solution to the monopoly fixed price. To determine monopoly real-time prices, plug in (B11) and (B12) in each other, which yields

$$\begin{aligned}
p_l &= \frac{a + w_l}{2}, \\
p_h &= \frac{a + w_h}{2}.
\end{aligned}$$